predictive CI is close to 0, indicating that we don't have any uncertainty about its value. This makes sense; after all, we already know what that value is from the training set.

SIDEBAR **Noisy function evaluations**

While not the case here, it's possible that the labels of the data points in our data set are noisy. This could happen in many situations in the real world where the process of observing data is corrupted by noise. In these cases, we can further specify the noise level with the GP and the CI at the observed data points will not collapse to 0 but instead to the specified noise level. This goes to show the flexibility that modeling with GPs offers.

This ability to assign a number to our level of uncertainty, which is called *uncertainty quantification*, is quite useful in any high-risk decision-making procedure such as Bayesian optimization. Imagine again the scenario in Table 1.1 where we tune the number of layers in our neural net, and we only have time to try out one more model. Let's say after being trained on those data points, a GP predicts that 25 layers will give a mean accuracy of 0.85 and the corresponding 95% CI is (0.81, 0.89). On the other hand, with 15 layers, the GP predicts that our accuracy will also be 0.85 on average, but the 95% CI is (0.84, 0.86). Here, it's quite reasonable for us to prefer 15 layers, even though both numbers have the same expected value. This is because we are more *certain* that 15 will give us a good result.

To be clear, a GP does not make any decision for us, but it does offer us a means to do so with its probabilistic predictions. Decision-making is left to the second part of the Bayesian optimization framework, the policy.

1.2.2 Making decisions with a Bayesian optimization policy

In addition to a GP as a predictive model, we also need in Bayesian optimization a decision-making procedure, which we explore in this subsection. This is the second component in Bayesian optimization, which takes in the predictions made by the GP model and reasons about how to best evaluate the objective function, so that the optimum may be located efficiently.

Remember from above, a prediction with a (0.84, 0.86) CI is considered to be better than one with (0.81, 0.89), especially if we only have one more try. This is because the former is more of a sure thing that is guaranteed to get us a good result. How should we make this decision in a more general case, where the two points might have different predictive means and different predictive levels of uncertainty?

This is exactly what a Bayesian optimization *policy* will help us do: quantifying the usefulness of a point given its predictive probability distribution. The job of a policy is to take in the GP model, which represents our belief about the objective function, and assign each data point with

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